

ABHISHEK SHANKAR

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Work Experience

RhythmX AI

December 2024 – Present

AI Solutions Engineer

Palo Alto, CA

- Integrated HL7 and FHIR data streams into AI workflows, enabling cross-domain clinical data access for AI features while enforcing HIPAA compliance across design, development, and production deployment
- Built and deployed RAG pipelines and multi-agent GenAI workflows for patient recommendation engines, clinical chat, and chart summarisation tools – owning the full lifecycle from design through DEV, staging, and PROD release
- Served as embedded technical partner to clinical stakeholders, debugging and optimising a live conversational AI copilot in production, resolving real-time inference issues directly impacting clinical workflows
- Scaled GenAI reasoning agents from 4 to 200+ medical conditions by optimising prompt orchestration, LLM pipelines, and GPU containers on Azure; reduced end-to-end inference latency by 50% through MongoDB query profiling and Temporal-based Batch API orchestration across 200,000+ requests

Boehringer Ingelheim

July 2023 – December 2023

Data Scientist Co-op

Ridgefield, CT

- Developed GraphRAG knowledge extraction pipeline using Stardog and LLMs, automating NLP-driven analysis of 1,000+ supply chain PDFs for executive reporting; delivered real-time GTM dashboards for 10+ drug launches via AWS CDK and Databricks
- Created 25+ Python and PySpark tutorial notebooks standardising AI development practices and accelerating enterprise Databricks adoption across cross-functional teams

Adqvest Capital Pvt. Ltd.

April 2020 – June 2022

Data & Software Engineer

Chennai, India

- Deployed 100+ Python scrapers and AWS RDS infrastructure to process gigabytes of alternative financial data; built NLP sentiment pipelines using SpaCy and PageRank algorithms, retaining 5+ clients worth \$2M+ AUM
- Architected 10+ statistical dashboards and REST API integrations enabling data-driven capital allocation decisions across complex financial portfolios

Projects

SupplyChainInsights – GraphRAG Knowledge Graph Platform ([GitHub](#) & [Medium](#)) May 2024 – August 2024

Python, Neo4j, Cypher, LangChain, OpenAI, RAGAS, Streamlit

- Featured in Neo4j's *This Week in Neo4j* newsletter – built an end-to-end RAG and multi-agent platform on Neo4j modelling 1,000+ supply chain entities; fine-tuned Llama 3 and Phi-3 for NL-to-Cypher query conversion
- Evaluated system quality using RAGAS (Faithfulness, Answer Relevancy); applied Graph Data Science similarity algorithms with OpenAI vector search to surface entity-level recommendations from structured clinical supply data

Parallel Image Captioning with PyTorch DDP ([GitHub](#))

January 2024 – May 2024

Python, PyTorch, CUDA, Multi-GPU

- Implemented distributed data-parallel training on COCO dataset using PyTorch DDP, achieving 3x faster convergence across 4 GPUs with mixed precision optimization
- Built encoder-decoder architecture with attention mechanisms, optimizing memory usage through parallel data loading for production-scale image captioning inference

Education

Northeastern University

August 2024

Master of Science in Information Systems (Data Science)

Boston, MA

SRM Institute of Science and Technology

May 2019

Bachelor of Technology in Mechanical Engineering

Tamil Nadu, India

Skills

LLM & GenAI: RAG (Vector & Graph), Multi-Agent Workflows, Prompt Engineering, LLM Evaluation, LangChain, MCP, Temporal Agentic Frameworks, OpenAI, Claude

Healthcare Data: HL7 v2, FHIR, REST APIs, HIPAA-Compliant Pipelines, Clinical Workflow Integration

Programming & ML: Python (Expert), SQL, PySpark, JavaScript, PyTorch, TensorFlow, Scikit-Learn

Cloud & DevOps: Azure (AKS, App Services), AWS (CDK, RDS, Athena), GCP, Docker, Kubernetes, CI/CD, Git

Databases: MongoDB, PostgreSQL, Neo4j, Stardog, ChromaDB, Clickhouse